M.Sc. Data and Knowledge Engineering

Module Handbook

Otto-von-Guericke-Universität Magdeburg · Faculty of Computer Science (FIN)

Valid from: Summer Semester 2026 (ab Sommer 2026)

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Note: The German version is legally binding. This document is compiled from the official FIN Bookstack pages.

1. Programme Overview

The Master's programme in Data and Knowledge Engineering (DKE) at OvGU Magdeburg delivers in-depth knowledge in Data Science, focusing on the engineering of data and knowledge systems. The programme equips graduates with skills to transform passive data into exploitable knowledge, covering representation, management, and understanding of data and knowledge assets.

Degree	Master of Science (M.Sc.)
Duration	4 Semesters (120 ECTS)
Language	English (German also possible)
Enrolment	Winter and Summer Semester
Admission	No restricted admission (NC-frei)
Entry Req.	B.Sc. in CS or related field, min. GPA 2.3 (German scale); 1 Databases course; C1 English or Germa
Faculty	Fakultät für Informatik (FIN), OvGU Magdeburg
Last Updated	June 2026 (Bookstack, ab Sommer 2026 version)

3. Module Descriptions by Area

A. Fundamentals of Data Science

Data Mining I – Introduction to Data Mining	
Module-ID: FIN-INF-120500	Semester: Summer | ECTS: 6
Foundation module on data mining principles, algorithms, and applications.	

Data and Knowledge Engineering	
Module-ID: FIN-INF-120500	Semester: Summer | ECTS: 6
Core methods for engineering data and knowledge systems.	

Introduction to Simulation	
Module-ID: FIN-INF-120345	Semester: Winter | ECTS: 6
Principles of simulation modelling and analysis.	

Machine Learning	
Module-ID: FIN-INF-120517	Semester: Winter | ECTS: 6
Statistical learning theory, supervised/unsupervised methods.	

Principles and Practices of Scientific Work	
Module-ID: FIN-INF-120XXX	Semester: Summer | ECTS: 3
Academic writing, research methodology, and soft skills for M.Sc. students.	

B. Learning Methods & Models for Data Science

Advanced Interactive Information Organization	
Module-ID: FIN-INF-120490	Semester: Winter | ECTS: 6
Methods for advanced information retrieval and interactive exploration.	

Advanced Topics in Deep Learning (Master)	
Module-ID: FIN-INF-120496	Semester: Every | ECTS: 6
State-of-the-art deep learning architectures and training techniques.	

Applied Discrete Modelling	
Module-ID: FIN-INF-120279	Semester: Winter | ECTS: 6
Graph-based and discrete mathematical models for real-world data problems.	

Data Mining II – Advanced Topics	
Module-ID: FIN-INF-120XXX	Semester: Winter | ECTS: 6
Advanced topics: ensemble methods, stream mining, pattern recognition.	

Human-Centred NLP	
Module-ID: FIN-INF-120XXX	Semester: Winter | ECTS: 6
Human-centred approaches to natural language processing systems.	

Introduction to Deep Learning

Module-ID: FIN-INF-120521	Semester: Summer | ECTS: 6

Neural networks fundamentals, backpropagation, CNNs, RNNs.

Introduction to ML Safety

Module-ID: FIN-INF-120532	Semester: Summer | ECTS: 3

Safety, robustness, and fairness in machine learning pipelines.

Learning Generative Models

Module-ID: FIN-INF-120522	Semester: Winter | ECTS: 6

VAEs, GANs, diffusion models, and generative AI evaluation.

Recommenders

Module-ID: FIN-INF-120XXX	Semester: Summer | ECTS: 6

Collaborative filtering, content-based, and hybrid recommendation systems.

Seminar 'Advanced Topics of KMD'

Module-ID: FIN-INF-120330	Semester: Every | ECTS: 3

Research seminar on current topics in knowledge and data management.

Swarm Intelligence

Module-ID: FIN-INF-102419	Semester: Winter | ECTS: 6

Bio-inspired optimisation: ant colony, particle swarm, evolutionary algorithms.

Eudaimonic Interaction Design

Module-ID: FIN-INF-120498	Semester: Winter | ECTS: 3

Design principles for meaningful and well-being-oriented user interfaces.

Individual Scientific Project (ISP)

Module-ID: FIN-INF-120301	Semester: Every | ECTS: 9

Self-directed research project under supervisor guidance.

C. Data Processing for Data Science

Advanced Database Models

Module-ID: FIN-INF-103805	Semester: Summer | ECTS: 6

NoSQL, graph databases, document stores, and polyglot persistence.

Advanced Topics in Databases

Module-ID: FIN-INF-120277	Semester: Summer | ECTS: 6

Query optimisation, transaction management, modern storage systems.

Introduction to Data Warehousing

Module-ID: FIN-INF-102808	Semester: Winter | ECTS: 6

ETL pipelines, star/snowflake schemas, OLAP, and BI tools.

Distributed Data Management

Module-ID: FIN-INF-102404	Semester: Winter | ECTS: 6
Distributed databases, CAP theorem, consistency models, replication.	

In-Memory and Cloud Technologies

Module-ID: FIN-INF-120506	Semester: Winter | ECTS: 6
In-memory DBMS, cloud-native data processing (AWS, Azure, GCP).	

Parallel Storage Systems

Module-ID: FIN-INF-120480	Semester: Summer | ECTS: 6
Parallel file systems, storage architectures for large-scale data.	

Transaction Processing

Module-ID: FIN-INF-103202	Semester: Winter | ECTS: 6
ACID properties, concurrency control, recovery protocols.	

VLBA – Cloud DevOps Technologies

Module-ID: FIN-INF-120478	Semester: Winter | ECTS: 6
DevOps, CI/CD, Kubernetes, containerisation for data engineering.	

Datenbanken 2 (DB Implementation Techniques)

Module-ID: FIN-INF-102XXX	Semester: Summer | ECTS: 6
Internal database implementation: indexing, buffer management, query plans.	

Geometric Data Structures

Module-ID: FIN-INF-102206	Semester: Summer | ECTS: 6
Spatial indexing, R-trees, k-d trees, geometric algorithms.	

D. Applied Data Science (Electives)

Data Science with Python

Module-ID: FIN-INF-120513	Semester: Winter | ECTS: 6
Practical data science: pandas, scikit-learn, visualisation, pipelines.	

MLOps for Small Language Model Applications

Module-ID: FIN-INF-120515	Semester: Winter | ECTS: 6
Model deployment, monitoring, fine-tuning, and lifecycle management.	

Scientific Writing

Module-ID: FIN-INF-120511	Semester: Winter | ECTS: 3
Academic writing skills, paper structure, LaTeX, peer review.	

IT-Security of Cyber-Physical Systems

Module-ID: FIN-INF-120457	Semester: Winter | ECTS: 6
Security in IoT/CPS environments, attack models, countermeasures.	

Visual Analytics

Module-ID: FIN-INF-120524	Semester: Summer | ECTS: 6

Interactive visual exploration of large datasets.

Visual Analytics in Health Care

Module-ID: FIN-INF-120476	Semester: Winter | ECTS: 6

Medical imaging, patient data analytics, health informatics.

Visualization

Module-ID: FIN-INF-102812	Semester: Winter | ECTS: 6

Information visualisation principles, D3.js, perception and cognition.

Narrative Visualization

Module-ID: FIN-INF-120523	Semester: Summer | ECTS: 3

Storytelling with data, interactive narratives, infographic design.

Human-Centred Approaches and Technologies (HCAT)

Module-ID: FIN-INF-120XXX	Semester: Winter | ECTS: 6

Usability engineering, user studies, accessibility.

Scrum-in-Practice

Module-ID: FIN-INF-120438	Semester: Winter | ECTS: 3

Agile project management, sprint planning, retrospectives.

Scientific Teamproject / KMD Teamproject

Module-ID: FIN-INF-120482	Semester: Every | ECTS: 6

Collaborative research project with industry or academic partners.

Seminar on Advanced Topics – Predictive Maintenance

Module-ID: FIN-INF-120XXX	Semester: Every | ECTS: 3

Research seminar: ML for predictive maintenance in industrial systems.

AMS Lab Projects

Module-ID: FIN-INF-120503	Semester: Every | ECTS: 6

Hands-on lab projects in autonomous/mobile systems.

Music Information Retrieval

Module-ID: FIN-INF-120516	Semester: Winter | ECTS: 6

Audio features, beat detection, genre classification, MIR systems.

Clean Code Development

Module-ID: FIN-INF-120432	Semester: Winter | ECTS: 3

Software craftsmanship, refactoring, design patterns, code review.

VLBA 1: Systemarchitectures

Module-ID: FIN-INF-120252	Semester: Winter | ECTS: 6

Large-scale system architectures: microservices, event-driven design.

User Experience Design Sprint	
Module-ID: FIN-INF-120510	Semester: Winter | ECTS: 3
Rapid prototyping and UX research methodology.	

E. Master's Thesis

Masters Thesis (Masterarbeit)	
Module-ID: FIN-INF-999997	Semester: Every | ECTS: 30
Independent research thesis supervised by a FIN professor. Duration: 6 months. Requires written thesis + oral defence (Kolloquium).	

4. Important Regulations & Notes

Use of Generative AI in Theses and Examinations

The Faculty Council (Beschluss 019/25) has issued regulations on using generative AI for final theses and examination work. Students must declare any AI tool usage and adhere to current FIN guidelines. Consult the Bookstack page 'Nutzung von generativer KI' for the latest binding rules.

Examination Committee (Prüfungsausschuss)

Binding decisions and regulations from the examination committee are documented on the Bookstack page 'Regelungen des Prüfungsausschusses'. These include module recognition, special examination conditions, and appeals processes.

Student Representation

The Fachschaftsrat der Fakultät für Informatik (FaRaFIN) is the elected student council representing all FIN students. The Academic Club is a student-run initiative for international students at the faculty.

Source: bookstack.cs.ovgu.de/books/msc-data-and-knowledge-engineering-ab-sommer-2026-U5t | FIN Study Regulations: fin.ovgu.de | Compiled June 2026